

Exp No: 18: GENERATION OF PWM USING TIMER INTERRUPT

PROGRAM:

```
#include <reg51.h>

sbit PWM = P1^0;

unsigned char count = 0;

unsigned char duty = 100; // Change this value (0–200)

void timer0_ISR(void) interrupt 1
{
    TH0 = 0xFC; // Reload for 1ms delay
    TLO = 0x66;

    count++;

    if(count < duty)

        PWM = 1; // ON time
    else

        PWM = 0; // OFF time

    if(count >= 200) // Total period = 200ms
        count = 0;
}

void main()
{
    TMOD = 0x01; // Timer0 Mode 1 (16-bit)

    TH0 = 0xFC; // 1ms preload
    TLO = 0x66;

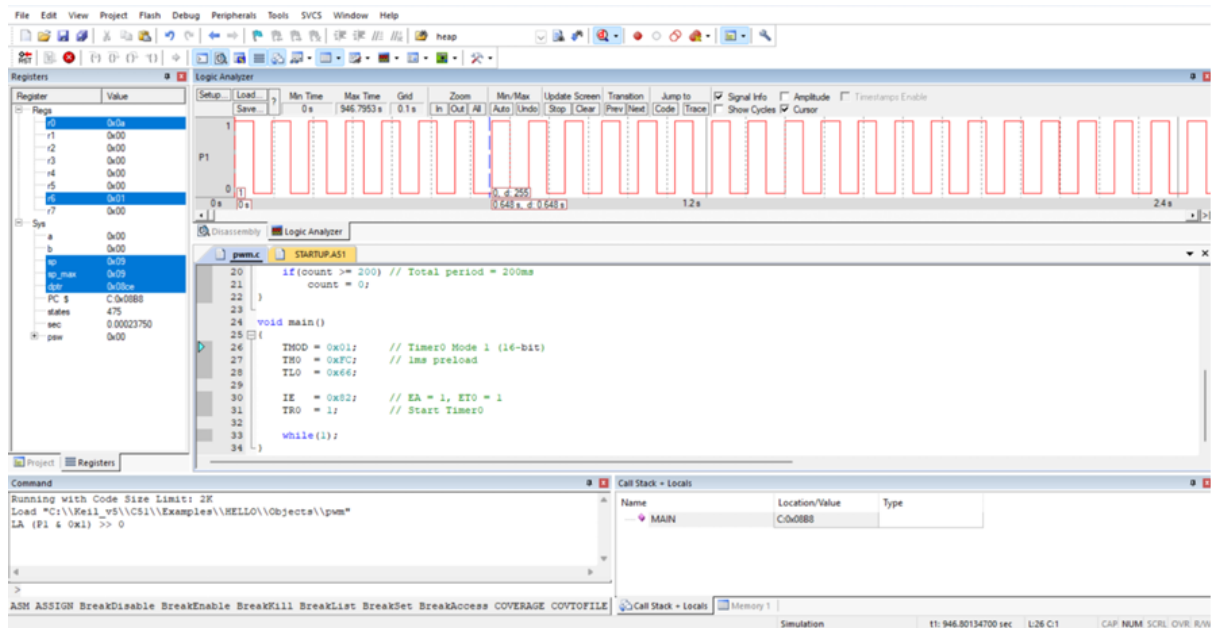
    IE = 0x82; // EA = 1, ET0 = 1

    TR0 = 1; // Start Timer0

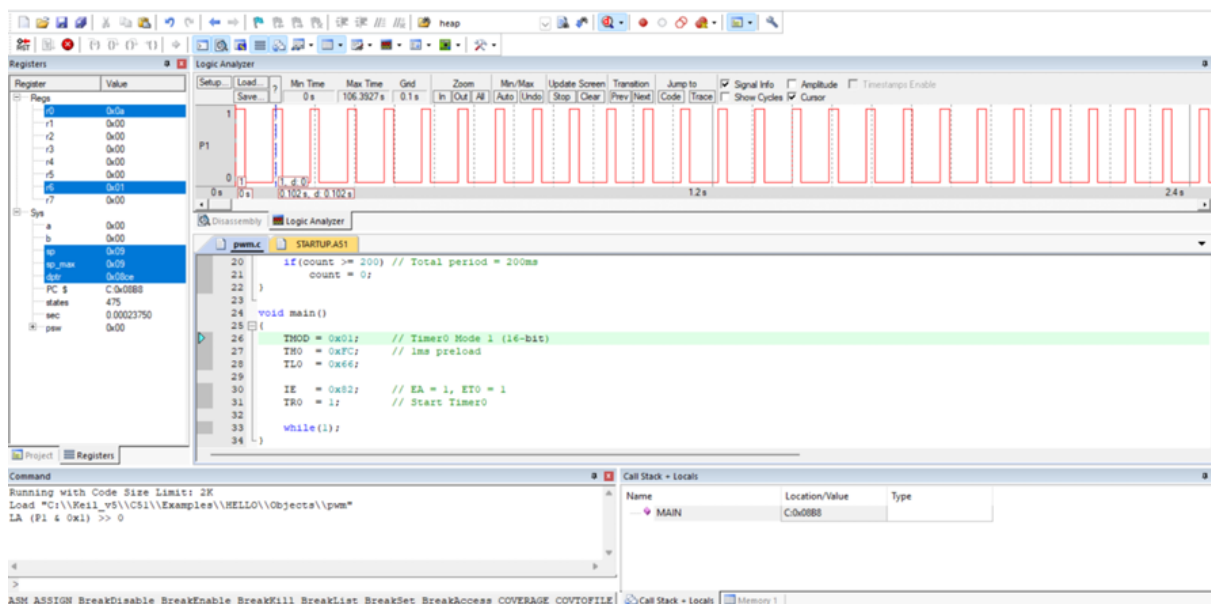
    while(1);
}
```

Output :

50% Duty cycle :



25% duty cycle :



75% duty cycle :

